THE SOLUTIONS

Notation:
(a) B - Bookwork
(b) E - New example
(c) A - New application

1. Consider two discrete random variables X and Y that have the joint probability mass function (pmf) represented by the joint probability table:

		Y	
		1	2
X	1	0.2	a
	2	0.4	b

a) Determine the marginal probabilities of X and Y in terms of a and b.

SOLUTION. For the first marginal: [2-E]

x	1	2
$P(X=x)$	0.2 + a	0.4 + b

The second is analogous: [1-E]

y	1	2
$P(Y=y)$	0.6	a + b

b) Is it possible to choose a and b such that X and Y are independent and the table describes a valid joint pmf? If yes, give a construction, if not, provide an argument.

SOLUTION. To ensure a valid joint pmf, we need to have $a+b+0.2+0.4=1$ and hence $b=0.4-a$. To further have independence, we need $P(X=x,Y=y)=P(X=x)\times P(Y=y)$ and hence: [1-E]

$$\begin{aligned} P(X=1,Y=1) &= 0.2 = P(X=1)\times P(Y=1) \\ &= (0.2+a)\times 0.6 = 0.12+0.6a \\ &\Longrightarrow a = \frac{0.08}{0.6} = \frac{2}{15} \end{aligned}$$

Hence:

$$b = 0.4 - \frac{2}{15} = \frac{2}{5} - \frac{2}{15} = \frac{6}{15} - \frac{2}{15} = \frac{4}{15}$$

We have then determined the only potential choice of a,b and need to check whether it satsifies the remaining factorizations of the pmf. For this choice of a and b: [2-E]

$$\begin{aligned} P(X=2,Y=1) &= 0.4 = P(X=2)\times P(Y=1) \\ &= (0.4+b)\times 0.6 = \left(\frac{6}{15}+\frac{4}{15}\right)\frac{3}{5} = \frac{10}{15}\times\frac{3}{5} = 0.4 \end{aligned}$$

Further:

$$P(X=1,Y=2) = a = \frac{2}{15} = P(X=1)\times P(Y=2)$$
$$= (0.2+a)\times(a+b) = \left(\frac{3}{15}+\frac{2}{15}\right)\left(\frac{2}{15}+\frac{4}{15}\right) = \frac{5}{15}\times\frac{6}{15} = \frac{2}{15}$$

Finally:

$$P(X=2,Y=2) = b = \frac{4}{15} = P(X=2)\times P(Y=2)$$
$$= (0.4+b)\times(a+b) = \left(\frac{6}{15}+\frac{4}{15}\right)\left(\frac{2}{15}+\frac{4}{15}\right) = \frac{10}{15}\times\frac{6}{15} = \frac{4}{15}$$

Hence, X and Y are independent for the choice $a = \frac{2}{15}, b = \frac{4}{15}$. [2-E]

c) **For the remainder of Question 1, we set $a = 0.3$ and $b = 0.1$.** Compute the probability that $X^2+Y \leq 3$, i.e., $P(X^2+Y \leq 3)$.

SOLUTION. Based on the choice $a = 0.3$ and $b = 0.1$, we specialize the joint pmf and marginals for the rest of the problem as:

		Y	
		1	2
X	1	0.2	0.3
	2	0.4	0.1

x	1	2
$P(X=x)$	0.5	0.5

y	1	2
$P(Y=y)$	0.6	0.4

We know that $X^2+Y \leq 3$ irrespective of Y whenever $X = 1$ and $X^2+Y > 3$ irrespective of Y whenever $X = 2$, and hence: [1-E]

$$P(X^2+Y \leq 3) = P(X=1) = 0.5$$

[2-E]

d) Compute the expectation of X, i.e. $E(X)$, and the expectation of Y, i.e. $E(Y)$.

SOLUTION. Making use of the marginal pmf's we computed in a), we have: [3-E]

$$E(X) = 0.5\times 1 + 0.5\times 2 = 1.5$$
$$E(Y) = 0.6\times 1 + 0.4\times 2 = 1.4$$

e) Compute the variance of X and the variance of Y, i.e. $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$, the covariance between X and Y, i.e. $\mathrm{Cov}(X,Y)$, and the correlation coefficient between X and Y, i.e. $\mathrm{Corr}(X,Y)$.

SOLUTION. We have $Var(X) = E(X^2) - (E(X))^2$ and hence: [2-E]

$$E(X^2) = 0.5\times 1 + 0.5\times 4 = 2.5 \Longrightarrow Var(X) = 2.5 - 1.5^2 = 2.5 - 2.25 = 0.25$$
$$E(Y^2) = 0.6\times 1 + 0.4\times 4 = 2.2 \Longrightarrow Var(X) = 2.2 - 1.4^2 = 2.5 - 2.25 = 0.24$$

For the covariance, we find: [2-E]

$$E(XY) = 0.2\times 1 + 0.3\times 2 + 0.4\times 2 + 0.1\times 4 = 2$$
$$\Longrightarrow Cov(XY) = E(XY) - E(X)E(Y) = 2 - 1.5\times 1.4 = -0.1$$

Finally, for the correlation: [1-E]

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)\times Var(Y)}} = \frac{-0.1}{\sqrt{0.25\times 0.24}} \approx -0.408$$

f) Are X and Y uncorrelated? Provide your reasoning.

SOLUTION. The correlation is not equal to zero, so the random variables are not uncorrelated. [2-E]

g) Compute the conditional probability mass function of X given Y, for each $Y = 1,2$.

SOLUTION. We have all the information we need in the joint and marginal pmf's to compute $P(X = x|Y = y) = P(X = x, Y = y)/P(Y = y)$: [2-E]

x	1	2
$P(X = x\|Y = 1)$	$\frac{0.2}{0.6} = 1/3$	$\frac{0.4}{0.6} = 2/3$

Analogously: [2-E]

x	1	2
$P(X = x\|Y = 2)$	$\frac{0.3}{0.4} = 3/4$	$\frac{0.1}{0.4} = 1/4$

h) Compute the conditional expectation of X given Y, for each $Y = 1,2$.

SOLUTION. Using the result from the previous question: [3-E]

$$E(X|Y = 1) = 1/3\times 1 + 2/3\times 2 = 5/3$$
$$E(X|Y = 2) = 3/4\times 1 + 1/4\times 2 = 5/4$$

i) By making use of (h), compute the expectation of X, i.e. $E(X)$. SOLUTION. Making use of the law of total expecation: [2-E]

$$E(X) = E(X|Y = 1)\times P(Y = 1) + E(X|Y = 2)\times P(Y = 2)$$
$$= 5/3\times 0.6 + 5/4\times 0.4 = 1 + 0.5 = 1.5$$

which conincides with the earlier result.

2. Consider a random variable Y with probability density function:

$$f_y(y) = \begin{cases} 2\lambda y e^{-\lambda y^2}, & \text{for } y \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the cumulative density function of Y.

SOLUTION. For $y \geq 0$, we have: [4-E]

$$F_y(y) = \int_{-\infty}^{y} f_y(u)du = \int_0^u 2\lambda u e^{-\lambda u^2} du = \left[-e^{-\lambda u^2}\right]_0^y = 1 - e^{-\lambda y^2}$$

For $y < 0$, it holds that $F_Y(y) = 0$ since $F_Y(y) = P(Y \leq y)$ and Y is never negative. [1-E]

b) We observe an independent and identically distributed (i.i.d.) sample of size n following the distribution for Y given above, denoted by $Y_1, Y_2, \ldots, Y_n$. Determine an expression for the maximum likelihood estimate of λ given this sample and verify that it is a maximum.

SOLUTION. For the likelihood, we have: [2-E]

$$\begin{aligned}\mathscr{L}_Y(\lambda) &= \prod_{i=1}^{n} f_Y(y_i) \\ &= \prod_{i=1}^{n} 2\lambda y_i e^{-\lambda y_i^2} \\ &= (2\lambda)^n \prod_{i=1}^{n} y_i e^{-\lambda y_i^2}\end{aligned}$$

As we have seen many times before, maximizing the log-likelihood is easier: [1-E]

$$\ell_Y(\lambda) = \ln \mathscr{L}_Y(\lambda) = n\ln(2) + n\ln(\lambda) + \sum_{i=1}^{n} \ln y_i - \lambda \sum_{n=1}^{n} y_i^2$$

Differentiating, and setting the derivative to zero, we find: [2-E]

$$\frac{n}{\hat{\lambda}} - \sum_{n=1}^{n} y_i^2 = 0 \Longleftrightarrow \hat{\lambda} = \frac{n}{\sum_{n=1}^{n} y_i^2}$$

Differentiating again, we note that the derivative is negative, and hence we have found a maximum: [1-E]

$$\frac{d^2}{d\lambda^2}\ell_Y(\lambda) = -\frac{n}{\lambda^2} < 0$$

c) Define $X = Y^2$, and determine the probability density function of X.

SOLUTION. Similar problems were covered in lectures and problem sheets, it is easier to work through the cumulative density function. By definition, X can only ever be non-negative, and for $x \geq 0$ we have: [2-E]

$$F_X(x) = P(X \leq x) = P(Y^2 \leq x) = P(Y \leq \sqrt{x}) = F_Y(\sqrt{x})$$

We found the cdf of X in part a) and can plug in: [1-E]

$$F_X(x) = F_Y(\sqrt{x}) = 1 - e^{-\lambda(\sqrt{x})^2} = 1 - e^{-\lambda x}$$

We can either directly recognize this as the exponential distribution or differentiate to find: [2-E]

$$f_X(x) = \frac{d}{dx}F_X(x) = \lambda e^{-\lambda}$$

d) We observe an i.i.d. sample of size n following the distribution for X derived in c), denoted by $X_1, X_2, \ldots, X_n$. Determine an expression for the method-of-moments estimate of λ given this sample.

SOLUTION. Since X is exponentially distributed, it holds that: [2-E]

$$E(X) = \frac{1}{\lambda}$$

and hence by the method of moments: [3-E]

$$\frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{\hat{\lambda}} \Longleftrightarrow \hat{\lambda} = \frac{n}{\sum_{n=1}^{n} x_i}$$